SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS - Final answers with no solution will get little

Exam Time: 90 minutes

Answer the following questions

Q1- Fit a straight-line to the data provided in the following Table: (7 marks)

xi	5	7	9	11	13
yi	6	4.5	3.4	2.3	1.4

The equation of the line is: .....

The value of *coefficient of determination*, r^2 is: .....

Q2- Compute the integration: $\int_2^4 \frac{1}{(x-1)^2} dx$, using Work to five significant figures. (10 marks)

Apply **Romberg integration** taking **n=1, n=2 & n=4** up to R(3 , 2).

Answer:

.....

Q3- Evaluate the following double integration: $\int_1^3 \int_1^3 (2x + y)\, dxdy$. (10 marks)

Take $dx = 0.5$ in the x-direction, $dy = 0.5$ in the y-direction. Apply the Trapezoidal rule in the x-direction and in the y-direction.

Answer:

.....

Q4- Maximize the function: $f(x, y) = xy + 4x - x^2 - y^2 - 4$ (10 marks)

Apply the Optimal steepest ascent method, use point (2, 2) as the initial estimate of the optimal solution. The optimum point after conducting one iteration is:

Answer:

.....

Q5- Consider the ordinary differential equation: $\frac{dy}{dx} = x^2 - 2y^2 + x$ (10 marks)

Given: $x_0 = 1,\ y_0 = 2, h = 0.5$

Use the Runge-Kutta method to the 4th order to find $y(1.5)$.

Answer:

.....

End of Questions

Q(1) straight-line The equation of the line is :- $y = 8.51 - 0.55x$

x_i	y_i
5	6
7	4.5
9	3.4
11	2.3
13	1.6

$R^2 = 0.9844 \Rightarrow 98.44\%$

sol $n = 5$

$\Sigma x^2 = 445$ $\quad \Sigma x = 45$ $\quad \bar{x} = \frac{\Sigma x}{n} = \frac{45}{5} = 9$

$\Sigma y^2 = 75.66$ $\quad \Sigma y = 17.8$ $\quad \bar{y} = \frac{\Sigma y}{n} = \frac{17.8}{5} = 3.56$

$\Sigma xy = 138.2$ $\quad \Sigma x^3 = 4725$

$\Sigma x^2 y = 1194.6$ $\quad \Sigma x^4 = 52789$

$y = A + Bx$ $\quad A = 8.51$ $\quad B = -0.55$ $\quad r = -0.99216$

$$a_1 = \frac{n \Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n \Sigma x_i^2 - (\Sigma x_i)^2} = \frac{(5)(138.2) - (45)(17.8)}{(5)(445) - (45)^2} = -0.55$$

$$a_0 = \bar{y} - a_1 \bar{x} = 3.56 - (-0.55)(9) = 8.51$$

كل الحلول

Q(2) $\int_2^4 \frac{1}{(x-1)^2} dx$ Apply Romberg integration $n = 1, n = 2$ & $n = 4$ up to (3, 2)

sol

Zero or	1st order	2nd order
R(1,0)		
R(2,0)	R(2,1)	
R(3,0)	R(3,1)	R(3,2)

$$R(2,1) = \frac{4^1 R(2,0) - R(1,0)}{4^1 - 1}$$

$= 0.7037$

$$R(3,1) = \frac{4^1 R(3,0) - R(2,0)}{4^1 - 1}$$

$= 0.671481$

$$R(3,2) = \frac{4^2 R(3,1) - R(2,1)}{4^2 - 1}$$

$= 0.66933$ //

* $R(1,0)$: $h = \frac{b-a}{n} = \frac{4-2}{1} = 2$

2, 4

$f(2) = 1$ $\quad f(4) = \frac{1}{9}$ $\quad I = \frac{h}{2}(f(2) + f(4))$

$= \frac{2}{2}(1 + \frac{1}{9}) = 1.111$

* $R(2,0)$: $h = \frac{b-a}{n} = \frac{2}{2} = 1$

2, 3, 4

$f(2) = 1$, $f(3) = 0.25$, $f(4) = \frac{1}{9}$

$I = \frac{1}{2}(1 + \frac{1}{9} + 2(0.25)) = 0.8056$

* $R(3,0)$: $h = \frac{b-a}{n} = \frac{2}{4} = \frac{1}{2} = 0.5$

2, 2.5, 3, 3.5, 4

$f(2) = 1$, $f(2.5) = \frac{4}{9}$, $f(3) = 0.25$, $f(3.5) = 0.16$, $f(4) = \frac{1}{9}$

$I = \frac{0.5}{2}(1 + \frac{1}{9} + 2(\frac{4}{9} + 0.25 + 0.16))$

$= 0.705$

محمد الحلوح

Q(3) $\int_1^3 \int_1^3 (2x+y)\,dx\,dy$ $dx = 0.5$ $dy = 0.5$ Apply Trapezoidal Rule x-direction and in the y-direction.

Sol

y / x	1	1.5	2	2.5	3
1	3	4	5	6	7
1.5	3.5	4.5	5.5	6.5	7.5
2	4	5	6	7	8
2.5	4.5	5.5	6.5	7.5	8.5
3	5	6	7	8	9

↓ y-axis

X-axis →

$h_1 = dx = 0.5$

$I_1 = \frac{0.5}{2}(3+7+2(4+5+6)) = 10$

$I_2 = \frac{0.5}{2}(3.5+7.5+2(4.5+5.5+6.5)) = 11$

$I_3 = \frac{0.5}{2}(4+8+2(5+6+7)) = 12$

$I_4 = \frac{0.5}{2}(4.5+8.5+2(5.5+6.5+7.5)) = 13$

$I_5 = \frac{0.5}{2}(5+9+2(6+7+8)) = 14$

$I = \frac{0.5}{2}(10+14+2(11+12+13)) = 24$ ✗

الحلول

Q(4) $f(x,y) = xy + 4x - x^2 - y^2 - 4$ Apply the optimal steepest ascent method.

Sol $(2,2)$ one iteration.

x_i ← $(2,2)$ → y_i

$(2,2)$

$\frac{df}{dx} = y + 4 - 2x = 2$

$\frac{df}{dy} = x - 2y = -2$

$\nabla f(2,2) = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$

$x_{i+1} = 2 + 2h$

$y_{i+1} = 2 - 2h$

$f(2+2h, 2-2h) = (2+2h)(2-2h) + 4(2+2h) - (2+2h)^2 - (2-2h)^2 - 4$

$4 - 4h + 4h - 4h^2 + 8 + 8h - (4 + 8h + 4h^2) - (4 - 8h + 4h^2) - 4$

$4 - 4h + 4h - 4h^2 + 8 + 8h - 4 - 8h - 4h^2 - 4 + 8h - 4h^2 - 4$

$-12h^2 + 8h = g(x)$

$g'(x) = -24h + 8$

$-24h + 8 = 0$

$\frac{24h}{24} = \frac{8}{24}$

$h = \frac{1}{3}$

0.333

$(\frac{8}{3}, \frac{4}{3})$

$(2.666, 1.334)$ ✗

الحلول

Q(5) O.D.E $\frac{dy}{dx} = x^2 - 2y^2 + x$ $y(1) = 2$

$x_0 = 1$, $y_0 = 2$, $h = 0.5$

Use the R.K method to the 4th order to find $y(1.5)$

$h = 1.5 - 1 = \underline{0.5}$

Sol

$K_1 = 0.5\, f(1, 2) = 0.5 * -6 = \underline{-3}$

$K_2 = 0.5\, f(1 + \frac{0.5}{2}, 2 + \frac{-3}{2}) = 0.5\, f(1.25, 0.5) = 0.5 * 2.3125$

$= \underline{1.15625}$

$K_3 = 0.5\, f(1.25, 2 + \frac{1.15625}{2}) = 0.5\, f(1.25, 2.5781125)$

$= 0.5 * -10.481 = \underline{-5.2405}$

$K_4 = 0.5\, f(1.5, -3.2405) = 0.5 * -17.2517$

$= \underline{-8.62585}$

$y(1.5) = 2 + \frac{1}{6}(-3 + 2(1.15625) + 2(-5.2405) + (-8.62585))$

$y(1.5) = -1.299058$ *

محمد الحلوح